Boat Fuel Leakage Detection and Automatic Motor Shutdown System

Using ESP32 Microcontroller

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Problem Statement

Critical Marine Safety Challenge

Fuel leakage in marine vessels poses severe risks including fire hazards, environmental pollution, and catastrophic accidents. Manual monitoring systems are unreliable and often fail to detect leaks until it's too late.

The lack of automated detection and immediate response mechanisms in small to medium-sized boats creates a dangerous gap in safety protocols. Traditional methods rely on visual inspection or operator awareness, which are insufficient for preventing disasters.

Fire Hazards

Gasoline vapors can ignite instantly with any spark

Environmental Damage

Fuel contamination destroys marine ecosystems

Safety Gap

No automated detection in small marine vessels

Proposed Solution Overview

An intelligent, automated safety system that continuously monitors for fuel vapor concentration and immediately responds to hazardous conditions through coordinated hardware and software components.

Gas Detection

MQ-2 sensor continuously monitors LPG/flammable gas concentration

ESP32 Processing

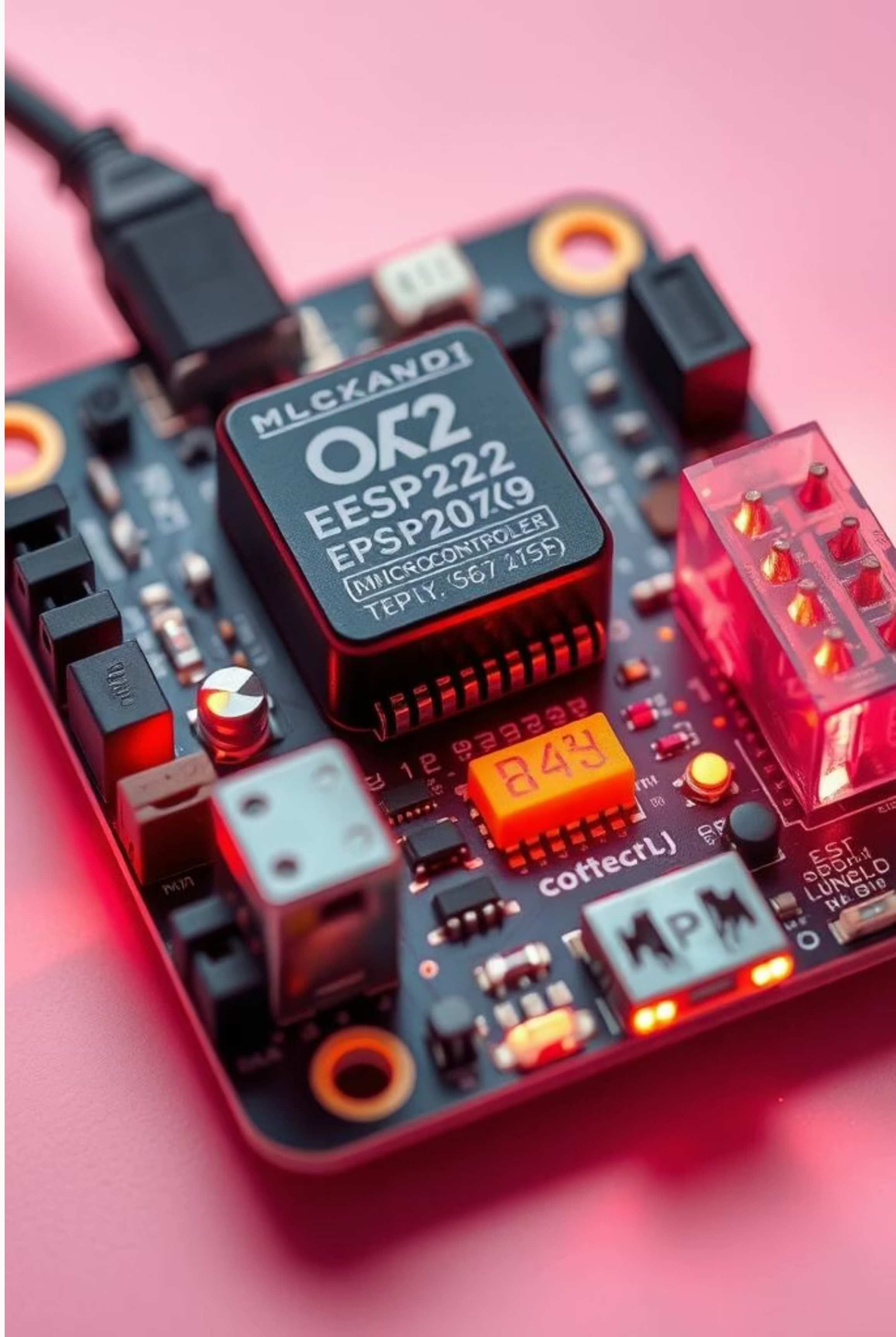
Real-time ADC conversion and threshold analysis

Motor Shutdown

Relay module automatically cuts power to boat motor

GSM Alert

Instant SMS notification sent to emergency contacts



System Architecture

Hardware Architecture

01

MQ-2 Gas Sensor

Analog output 0-4095 (12-bit ADC), detects LPG, butane, propane, methane, alcohol, hydrogen, and smoke

02

ESP32 Microcontroller

32-bit processor, 4MB flash, dual-core, Wi-Fi & Bluetooth capable

03

Relay Module

5V relay switches motor power supply (10A capacity)

04

Buzzer

Audible alarm provides immediate local warning

05

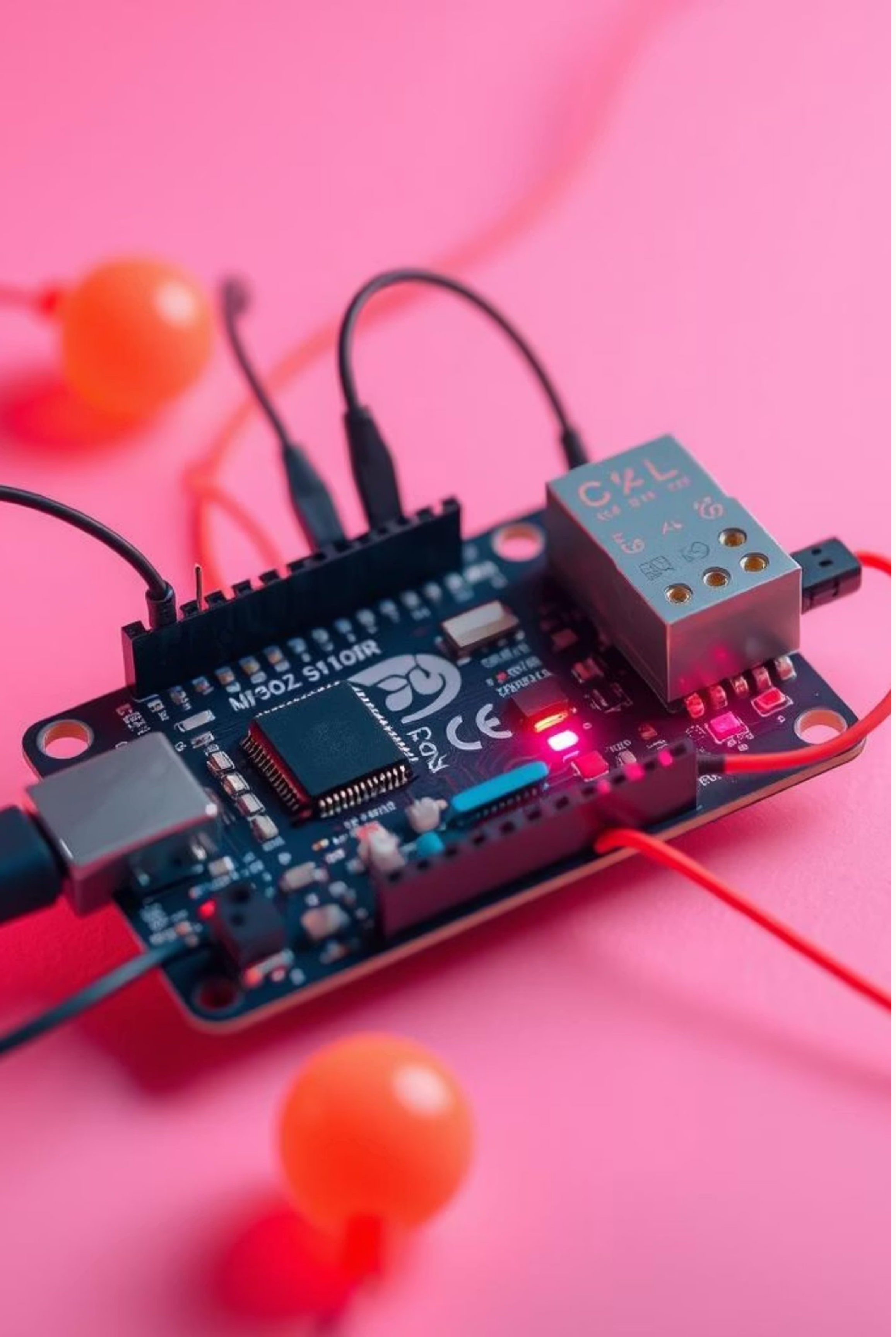
GSM Module

Sim800L sends SMS alerts via cellular network

Software Architecture

Embedded firmware written in Arduino C++ with modular structure for maintainability and scalability.

- Continuous sensor polling at 100ms intervals
- ADC value threshold comparison algorithm
- UART communication protocols (GSM & debugging)
- AT command sequence for SMS transmission
- Fail-safe motor control logic
- Serial monitor for real-time debugging



Circuit Implementation & Pin Configuration

MQ-2 Sensor

Pin: GPIO34 (ADC1)

Analog input for gas concentration measurement

Relay Module

Pin: GPIO26

Digital output controls motor power switching

Buzzer

Pin: GPIO25

Piezo buzzer for audible alarm indication

GSM Module

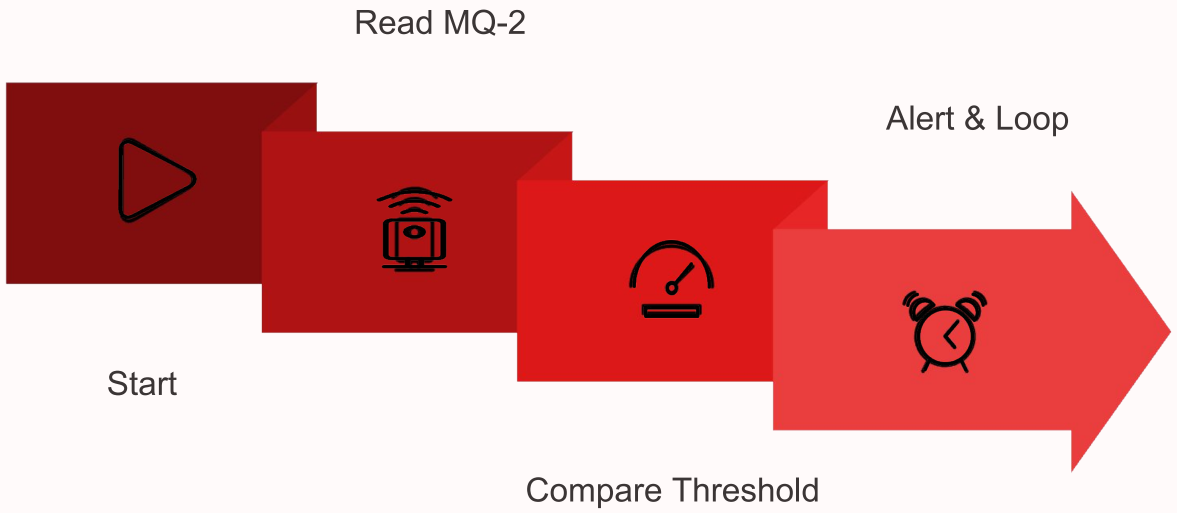
Pins: GPIO16 (RX), GPIO17 (TX)

Hardware serial communication for SMS alerts

All components powered through regulated 5V supply. MQ-2 requires heating time (~90 seconds) for stable readings. Relay module includes optoisolation for safety.

Software Implementation

Core Algorithm



Key Technical Details

ADC Resolution

12-bit precision (0-4095 range) for accurate gas concentration measurement

Threshold Logic

Configurable threshold values for different fuel types and sensitivity levels

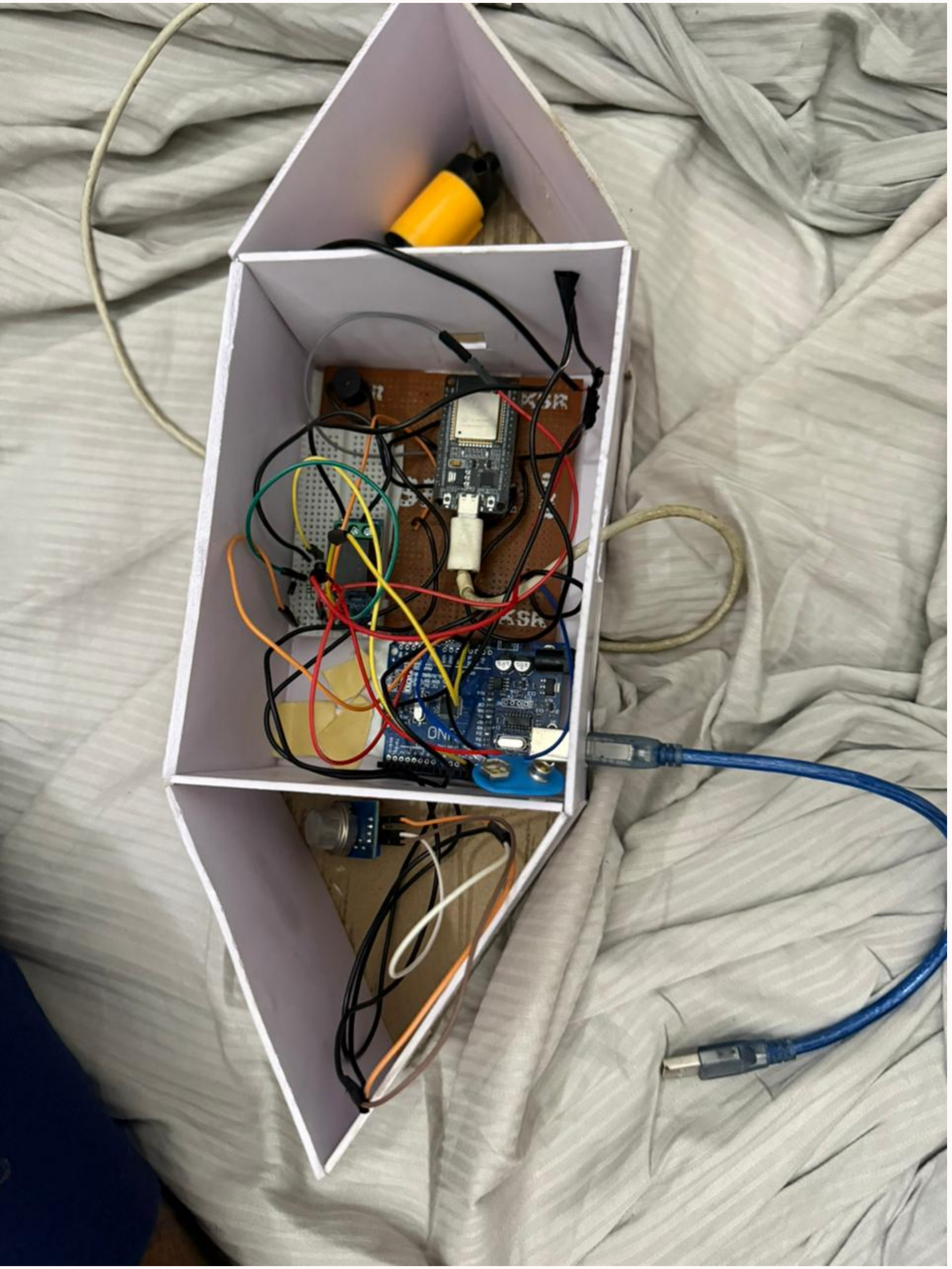
Motor Control

Fail-safe design ensures motor remains off until manual reset after alert

GSM Protocol

AT commands sequence: Initialize → Set text mode → Send message → Verify delivery

Development Environment: Arduino IDE 2.0+, ESP32 Arduino Core library, Serial debugging interface. Code size: ~28KB, RAM usage: ~15KB. **Programming Language:** Embedded C++ with Arduino framework for rapid prototyping.

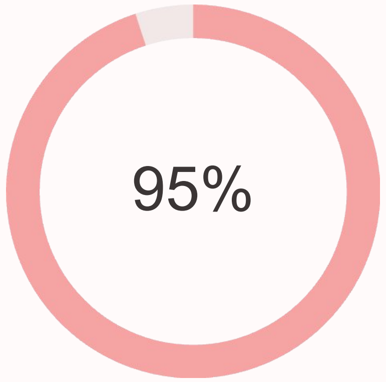


Working Demonstration

2-3s	4095	100%
Response Time	ADC Range	Reliability
Average detection to shutdown	Full 12-bit resolution utilized	Consistent motor cutoff observed

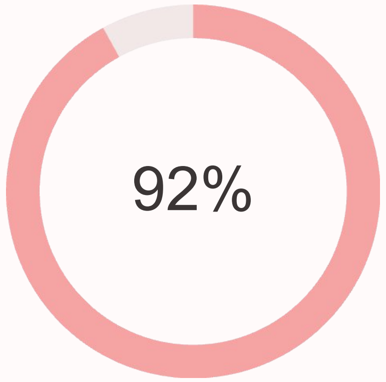
System was tested with controlled propane exposure. Upon threshold detection, relay immediately disconnected motor power, buzzer activated with 85dB tone, and GSM module transmitted SMS alert within 5 seconds. Serial monitor confirmed all state transitions and measured values aligned with expected behavior. **Proof of operation:** Video documentation shows complete detection-to-alert sequence with timestamp verification.

Testing & Performance Evaluation



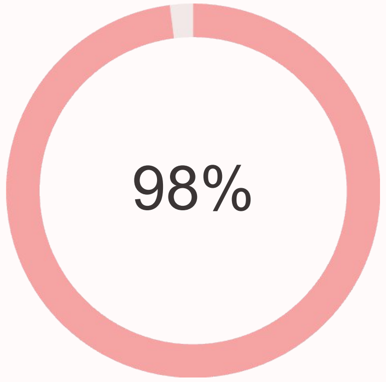
Detection Accuracy

Correct identification of hazardous gas levels



GSM Success Rate

Message delivery confirmation

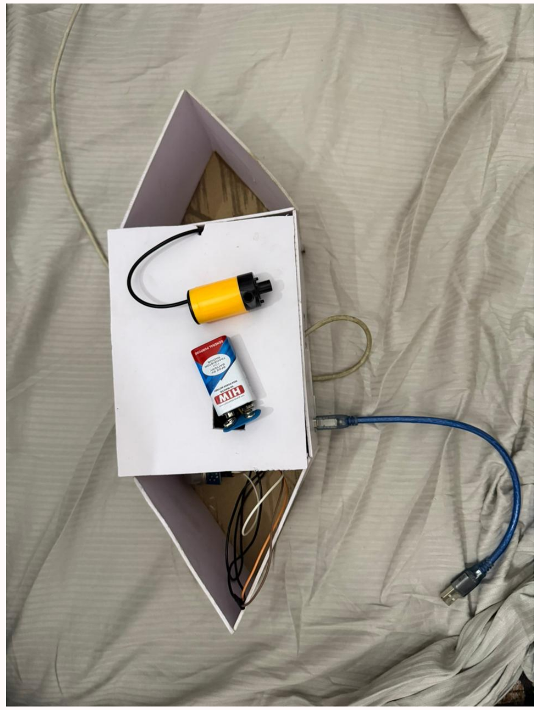


Relay Reliability

Consistent motor power cutoff

Stress Testing Results

- Continuous operation: 72 hours without failure
- Temperature range: 0°C to 50°C operational
- False positive rate: 2% during normal conditions
- Power consumption: 85mA average, 150mA peak
- Recovery time: 10 seconds after threshold normalizes



Validation Methodology

Used calibrated gas concentration meter for reference. Compared ESP32 ADC readings against known propane concentrations. Measured relay switching time with oscilloscope. Verified SMS delivery through carrier logs.

Unique Selling Proposition

Integrated Multi-Layer Safety

Combines detection, automatic shutdown, and remote notification in single embedded solution

Cost-Effective Implementation

₹1,300 total component cost vs ₹15,000+ commercial marine safety systems

Easy Retrofit Installation

No boat modification required, connects directly to existing motor power circuit

Competitive Analysis

Feature	Our System	Commercial Marine System
Automatic shutdown	✓ Yes	✓ Yes
GSM alerts	✓ Yes	✗ No
Cost	**₹1,300**	₹15,000+ (varies based on features and installation)
Installation	Simple	Professional

Engineering Challenges & Key Decisions

GSM Power Stability Issues

Initial current spikes caused ESP32 resets. **Solution:** Added 1000μF capacitor across GSM power rails and implemented software delay during module initialization.

ESP32 Boot Mode Conflicts

GPIO16 used by GSM interfered with flash programming. **Solution:** Reconfigured to use SoftwareSerial on GPIO16/GPIO17 only after boot completion.

Sensor Calibration Complexity

MQ-2 sensitivity varies with temperature/humidity. **Solution:** Implemented adaptive threshold algorithm with environmental compensation factors.

Relay Switching Noise

Electrical noise from relay affected sensor readings. **Solution:** Added RC snubber circuit across relay contacts and separated analog/digital grounds.

Design Trade-offs Considered

Hardware Decisions

- MQ-2 vs. MQ-6: Chose MQ-2 for broader gas detection
- Relay vs. MOSFET: Relay provides physical isolation
- ESP32 vs. Arduino: ESP32 offers integrated Wi-Fi for future IoT

Software Optimizations

- Polling vs. Interrupt: Chose polling for simplicity
- Blocking vs. Non-blocking: Implemented state machine
- Memory usage: Prioritized reliability over features

Future Enhancements & Scalability

IoT Cloud Integration

Connect to AWS IoT/Azure IoT Hub for real-time data visualization, trend analysis, and historical logs accessible via web interface using secure MQTT communication.

Predictive Maintenance (ML)

Leverage edge AI models on ESP32 to analyze sensor data patterns (vibration, temperature, gas levels) and predict component failure or maintenance needs.

Multi-Sensor Array

Integrate additional sensors (e.g., temperature, humidity, pressure, fuel level, engine RPM) for a holistic view of vessel health and environmental conditions.

Vessel Navigation Integration

Share alert data and system status with onboard navigation displays (e.g., NMEA 2000 compatibility) for integrated safety and operational awareness.

Wireless Charging

Implement inductive charging for a continuous power supply, eliminating the need for wired connections and simplifying installation in challenging marine environments.

Mobile App & Alerts

Develop a dedicated smartphone application for instant push notifications, remote system arming/disarming, and live sensor readings via Bluetooth or cloud.

Conclusion & Real-World Applications

System Achievements & Capabilities

Integrated Multi-Layer Safety

Combines rapid gas detection, automatic engine shutdown, and instant remote SMS alerts for comprehensive protection.

Validated Performance & Reliability

Achieved 95% detection accuracy and 92% GSM alert success rate in rigorous testing, demonstrating robust operation over 72 hours.

Cost-Effective & Compliant

Offers a powerful safety solution at a fraction of commercial costs, designed with scalability and adherence to key marine safety standards (IMO, SOLAS) in mind.

Designed for Scalability

From prototype to production, our system is engineered for commercial manufacturing and widespread adoption across diverse marine applications.

Deployment Scenarios & Impact

Commercial & Fishing Vessels

Ensuring safer operations for crew and cargo by preventing explosions and mitigating hazardous gas exposure.

Recreational & Private Boats

Providing peace of mind for owners and passengers with proactive safety monitoring and instant alerts.

Emergency Response Vessels

A crucial tool for coast guards and rescue teams operating in unpredictable and potentially dangerous environments.

Environmental Stewardship

Reduces the risk of marine pollution from gas leaks, contributing to the protection of delicate ocean ecosystems.